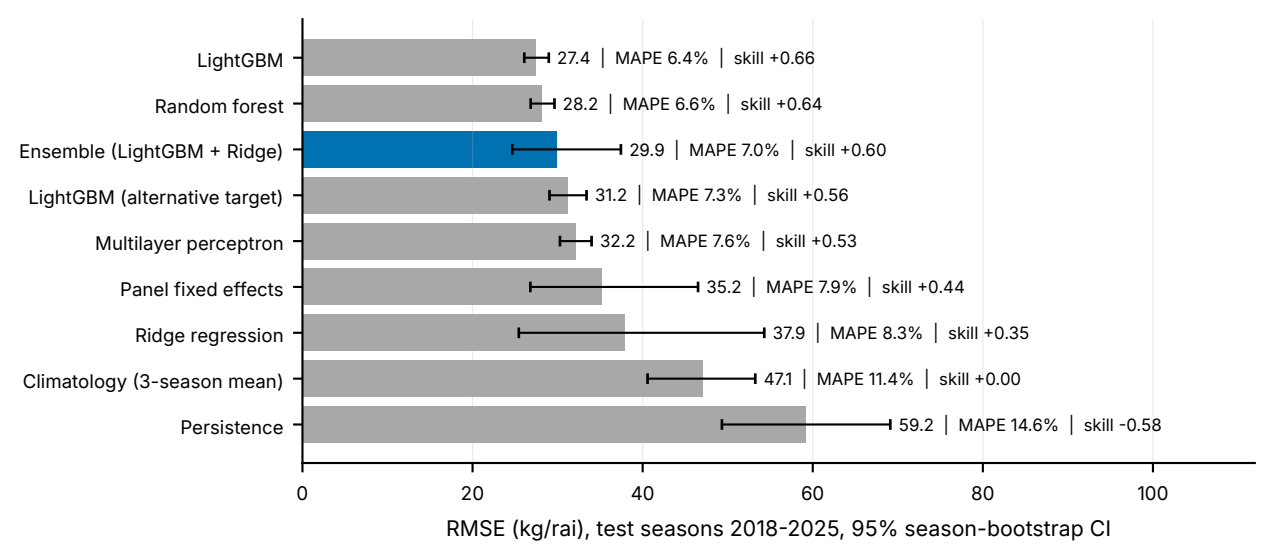
LightGBM
27.4 | MAPE 6.4% | skill +0.66
Random forest
28.2 | MAPE 6.6% | skill +0.64
Ensemble (LightGBM + Ridge)
29.9 | MAPE 7.0% | skill +0.60
LightGBM (alternative target)
31.2 | MAPE 7.3% | skill +0.56
Multilayer perceptron
32.2 | MAPE 7.6% | skill +0.53
Panel fixed effects
35.2 | MAPE 7.9% | skill +0.44
Ridge regression
37.9 | MAPE 8.3% | skill +0.35
Climatology (3-season mean)
47.1 | MAPE 11.4% | skill +0.00
Persistence
59.2 | MAPE 14.6% | skill -0.58
0
20
40
60
80
100
RMSE (kg/rai), test seasons 2018-2025, 95% season-bootstrap CI